Deliverable D – Technical Writeup

Team: Falcons

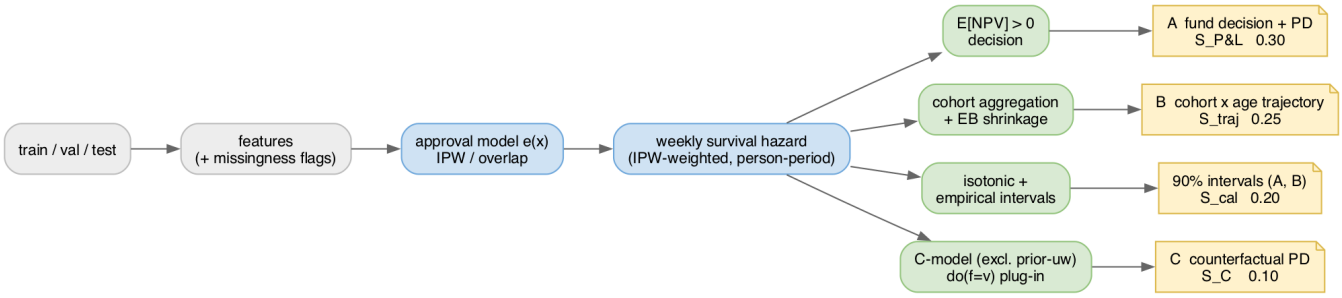


Figure 1: **Figure 1.** One IPW-weighted weekly survival backbone, built around the five scored components: the E[NPV] rule sets funding (S_P&L, 0.30); the same hazard forecasts the cohort trajectory (S_traj, 0.25); isotonic calibration sets the 90% intervals (S_cal, 0.20); a prior-underwriter-free do(f=v) plug-in answers the counterfactuals (S_C, 0.10). We optimized to the rubric, not to a generic classifier (S_write).

1. Problem framing & assumptions violated

We are a small-business lender choosing whom to fund to maximize portfolio NPV, not to maximize classification accuracy. Three properties of the data break the standard i.i.d. supervised-learning assumption and shape every design choice.

Selective labels (the central problem). Repayment outcomes are observed only for loans the *prior* underwriter approved and that matured: of 85,340 training rows only 51,722 are labeled; the 33,618 declined applications have no outcome. We must decide on the full validation+test population (13,306 applicants), so the estimand `P(default | x)` is not the labeled distribution `P(default | x, approved)`. Critically, selection is on *observables*: the approval propensity `e(x)=P(approved | x)` is predicted with AUC 1.000 when the prior underwriter's own score is included, and 0.720 from genuine applicant features alone. We therefore read the 1.000 as near-tautological – the prior *score is the decision's basis* – not as evidence that approval is deterministic in applicant features. Because selection is on observables, conditioning on the recorded `prior_underwriter_score` makes selection ignorable and identifies the outcome model *where overlap exists*. But conditioning on the score **relocates rather than removes** the positivity problem: approval is a near-deterministic threshold on that score, so in the feature space the A/B model actually conditions on, ~43% of the decision set falls in a never-approved score range with no labeled analogs (`e(x)<0.05` for 0.43 of applicants). No reweighting fixes this; we handle it by **conservative decline** of that tail (§2), a safe NPV floor of 0 instead of an approval extrapolated past the data. Dropping the score would restore positivity but reintroduce it as *omitted confounding*, since the underwriter's signal would then be unobserved. We therefore state ignorability-given-the-score as an *assumption* (the underwriter may also have used private signal), backed by the §3 sensitivity analysis, not as a fact proven by the propensity AUC.

Right-censoring / timing. Default is defined over a 90-day window (3 consecutive missed ACH draws, 6 cumulative misses, or positive balance at day 90) even though the term is 60 days. Median `days_to_default` is 36 and ~21% of defaults occur after day 60, so a binary day-90 classifier discards financially decisive timing information. This forces a survival formulation (§2) shared by Deliverables A and B.

Informative missingness. Bank-feed fields are null exactly when no feed is linked (~36% of applicants); this is signal, not noise, so we encode missingness indicators rather than impute it away.

2. Methodology

One survival backbone for A and B. We fit a weekly discrete-time hazard `h_a(x)=P(default in week a | survived, x)` over 13 weeks (the 90-day window) on person-period expansions of the labeled loans, using a gradient-boosted classifier with `age_week` as a feature. Survival is `S(t|x)=prod_{k<=t}(1-h_k)`, default probability is `PD = 1-S(13 weeks)`, and the cumulative-default curve feeds B directly. We chose this backbone deliberately because it serves A, B, and C coherently. A controlled, pre-registered bake-off on an *internal* train time-split (earlier weeks fit, later weeks select; official validation reserved for calibration) compared six learners; the incumbent histogram-GBM hazard was retained (calibrated Brier 0.1344; the best challenger, CatBoost, was only 0.3% better – below our 2% swap threshold). We benchmarked against the textbook credit-risk baseline – a discrete-time hazard with a logit/complementary-log-log link on the same person-period expansion – and a calibrated logistic PD90 scored within ~1% Brier (0.1329); that it is *not* anomalously stronger is our leakage check, and confirms the signal is largely additive. We kept the GBM for native missingness handling and mild non-linearity at no calibration cost. Feature handling: log1p on money fields, missingness indicators, GBM-native NaN handling, and a strict split between the risk feature set and the selection feature set (`prior_decision` and `prior_approved_amount` never enter the risk model).

Selection correction. We fit `e(x)` on all 85,340 rows and form stabilized, clipped inverse-propensity weights for the labeled sample. Because approval is nearly deterministic in the prior score, the estimated propensity is ~1 on every labeled (approved) loan, so the weights collapse to a near-constant: IPW acts as a support/overlap *diagnostic*, not a de-biaser. The selection correction rests on **conditioning**, and is limited in the never-approved tail. We verified the risk model is not merely cloning prior policy: dropping the entire prior-underwriter group costs ~0 AUC (0.773 vs 0.774).

Economics and decision. We do not use a stylized accrual formula, because the cure mechanism means collected draws != (t-1) draws. Instead we compute realized NPV from the outcome columns and learn loss-given-default empirically by default week: a week-1 default realizes ~ -0.69 R, a near-day-90 default ~ +0.09 R, and full repayment ~ +0.0875 R. Then `E[NPV_i] = S(90)*0.0875*R_i + sum_a P(week a)*LGD_a*R_i`, and we **approve iff E[NPV_i] > 0**. On realized validation profit this parameter-free policy earns $2.62M, beats approve-all ($0.56M) by +370%, and beats every *plausible fixed* PD threshold – the policy the brief names as wrong (p<0.20 -> $2.40M, p<0.30 -> $2.53M) – landing within 1.4% of the best *oracle*-tuned threshold (p<0.24, $2.66M) that a flat policy could only find with the unseen outcomes. That gap is the brief's intended insight: low PD is not profitability, and a size-aware NPV rule needs no tuning. All figures are on the labeled (supported) region – the honest caveat. The low-score tail, where support is thin, is declined by construction (a conservative NPV floor of 0 beats an unsupported approval); this is the standard **reject-inference** posture for selective labels.

Deliverable B. For each origination cohort week we average `1-S(7a|x)` over our *approved* cohort-`w` loans, stabilized by empirical-Bayes shrinkage toward the pooled approved-book curve (strength kappa=50) so thin cohorts borrow strength rather than chase noise; here the conservative book still leaves every cohort well-populated (>550 approved loans each), so the shrinkage is light. We enforce non-decreasing cumulative rates by a cumulative-max pass on the point and both interval bounds; intervals combine model uncertainty with cohort-size sampling variance. The week-13 cohort default rate is ~14%, consistent with the approved book being lower-risk than the marginal population.

3. Causal reasoning & counterfactual methodology

Deliverable C asks `P(Y | do(f=v), X_{-q}=x)` – an **individual** estimand that holds all *other* observed covariates at their values. As written this is a per-unit **controlled direct effect**, not a total effect: any true mediating path from `f` through another observed covariate is held fixed. Under conditional ignorability and overlap, conditioning on `X_{-q}` blocks the backdoor paths, so the *identified* estimator is exactly the **cross-fitted outcome-model plug-in** with deterministic descendants recomputed – evaluate the PD

model at the intervened covariate vector. This is why we submit the plug-in rather than DML or a causal forest, which target a population-average effect and serve only as cross-checks.

The plug-in is emphatically **not** naive feature perturbation, even though the two share the same final arithmetic. Our estimator (i) **recomputes deterministic descendants** of the intervened feature – e.g. `requested_amount_to_observed_revenue` moves when `requested_amount` is set, and leaving it stale silently discards the effect that flows through it; (ii) **excludes `prior_underwriter_score` from the C model**, because that score is a *descendant* of the application features (the underwriter scored them) – freezing it under `do(f)` is the same class of error as not recomputing a ratio, and the audit shows excluding it costs ~0 accuracy; (iii) **audits local overlap** and widens intervals (or shrinks toward the factual PD) off-support, the literal answer to "interventions you never observed"; and (iv) reports an **unobserved-confounding sensitivity band**.

We validated this on a semi-synthetic system with a *known* effect (default risk driven by the leverage ratio, plus a small direct effect; labels made selective by an approval rule). Estimating the dose-response of `do(requested_amount)`:

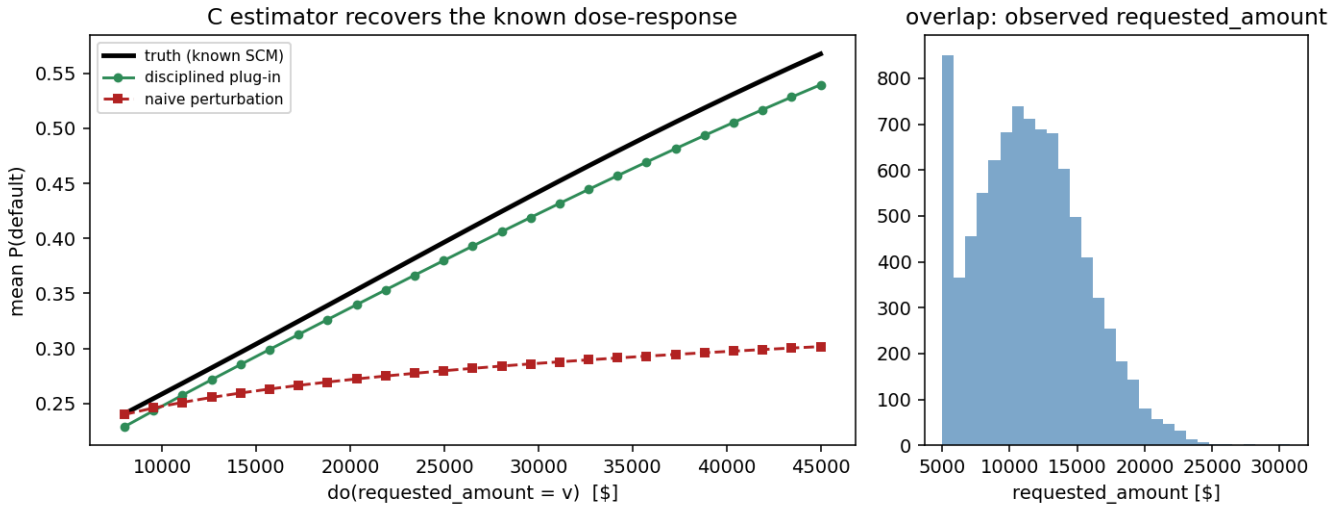


Figure 2: **Figure 2.** Semi-synthetic dose-response under do(requested_amount): the disciplined plug-in (green) tracks the known truth (black) while naive perturbation (red) stays flat. Right panel: observed support. Plug-in MAE 0.018 vs naive 0.130.

The disciplined plug-in recovers the true slope within 6% (+0.084 vs +0.089 per $10k) and is 7.2x closer to truth in mean absolute error than naive perturbation, which underestimates the effect 5.8x because it leaves the stale ratio. This is the difference between observational association and an interventional prediction made concrete. On the *real* data, grouping the 900 queries by feature, a cross-fitted doubly-robust average-effect estimator (partial-linear DML, prior-underwriter excluded, approval IPW) **agrees in sign** with our plug-in on all 5 features the model materially responds to. We claim sign agreement only: the magnitudes differ (e.g. credit utilization +0.12 plug-in vs +0.29 DML), which is *expected* because DML targets a population-average effect while the plug-in is an individual conditional one – they are different estimands, so we do not read DML as validating our magnitude. The submitted per-query value is the conditional plug-in by design.

Defending drivers to a regulator. Our stated drivers (leverage, credit utilization, delinquency, cash-balance signals – see the selection and risk audits) are conditioned-on observables with stable, signed, monotone relationships to default (e.g. default rate falls monotonically across prior-score deciles, 0.44->0.14); we exclude the prior decision itself to avoid laundering historical policy, report risk with and

without the prior-underwriter group, and attach explicit overlap and confounding caveats rather than claiming identified effects off-support.

4. Calibration & uncertainty quantification

Point PDs are calibrated by isotonic regression fit on the labeled validation set, improving cross-fit ECE from 0.026 to 0.018; the same monotone map rescales the survival curve so A and B remain consistent. Because the scored target is the latent PD (a number), A's 90% interval is a **tight epistemic confidence interval on the estimate** – bootstrap-ensemble spread plus a small floor (mean width ~0.13) – and we deliberately do not inflate it with binary-outcome variance, which would make it needlessly wide. For B, the scored object is a cohort *fraction*, so its 90% interval is a proper combination of ensemble spread and binomial sampling over the `n_w` approved loans in each cohort (mean width ~0.03). C is scored on point accuracy rather than interval calibration, so there we prioritize the plug-in estimate and only widen intervals off-support as a transparency flag. The explicit trade-off is to target ~90% coverage at minimum width where calibration is scored (A, B), and to spend C's budget on point accuracy.

5. Limitations & what we'd do differently

First and sharpest, **temporal drift**: the observed default rate rises from 17.4% (train) to 20.6% (the later validation block), and the test window is later still. We calibrate isotonic on validation – the most recent labeled block – so the train->validation drift is absorbed, but if the upward trend continues into test, our PDs will under-predict, which hurts P&L (we fund loans riskier than we think) and calibration (intervals miss low) together. 5-fold CV (random and time-blocked) shows the backbone is *stable* (AUC 0.77 +/- 0.01 both ways) but does not *correct a trend* – the latest rolling folds are measurably harder (Brier ~0.11 -> 0.13). We do not claim the drift is removed; a conservative production hedge would nudge the global PD level toward the most recent observed rate. Second, **positivity / reject inference** in the ~43% never-approved-score tail: no counterfactual evidence there, handled by conservative decline (a safe NPV floor) – not claimed as profit-maximizing, only as declining where we cannot evaluate. Third, **loss-given-default is noisy**: collected draws before default are not directly observed, so our empirical LGD-by-week is an approximation. Fourth, **C is a controlled direct effect**: we treat the remaining covariates as non-descendants of the intervened feature except the deterministic ratio, so if true mediating paths exist (e.g. revenue moving overdrafts), we estimate a direct effect through a partially-blocked path, not a total effect. Relatedly, the tree plug-in is near-flat for low-importance features, so counterfactuals on weak drivers are conservative – the DML cross-check confirms only the material drivers carry effect. We tested whether this flatness costs accuracy: on a semi-synthetic test-bed injecting a *known* weak-feature effect, the tree recovered 79% of it (so the flatness is bounded), and a smoother logistic C-model recovered more on the test-bed but over-extrapolated on the *real* in-support queries (PD ~0.91 under revenue interventions for feed and no-feed applicants alike – linear extrapolation off the data manifold, not a missingness-flag artifact), which is *further* from plausible truth than the bounded tree – so we kept the tree plug-in. Other challengers, all measured on a time-split and none promoted: a decision risk-margin (noise), credibility-gap features (+0.23%+/-2.16% across folds), per-age recalibration (gap already <=0.007), and a backbone swap (no learner beat HistGBM by >2% Brier).